COMP20008 REPORT
GROUP W10G1

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Executive Summary

This report analyses the factors impacting housing price changes over 2013-2023. Various suburbs were investigated, and by using the two data sets 'communities.csv' and 'Houses-by-suburbs.csv", we commenced with our exploratory data analysis. Visual tools such as bar charts and heatmaps were implemented, as well as calculating mutual information to develop a more thorough understanding of the relationships between the factors. An observation of these tools revealed that the factors relating to low socioeconomic standing, such as IRSD and number of households with an income of <$600/week had surprisingly low correlations to the change in housing price.

We then trained machine learning models, specifically linear regression and k-Nearest Neighbours (kNN) models to predict the change in housing prices. The linear regression model revealed that IRSD and change in housing price had a R^2 value of 0.0629, suggesting a very low linear correlation between the two. This led to the implementation of the kNN model, which is able to handle both linear and non-linear relationships. This was able to achieve an accuracy score of 0.75, however, only after omitting IRSD and housing prices in 2023 in the feature selection.

We thus recommend that the government consider IRSD more strongly when establishing housing prices.

1. Introduction

In this report, we will explore the factors that affect the change in housing prices from 2013-2023 and analyse how these changing prices may be related to the number of socioeconomically disadvantaged families in the suburb.

The central research question this report addresses is "How do housing prices changes from 2013-2023 affect the number of low-income families in suburbs, and what factors contribute to these changes?". Using the two data sets, 'communities.csv' and 'Houses-by-suburbs.csv', we seek to understand and analyse the data and factors that have driven changes in housing prices. Not only that, we will examine the relationship between the change in house prices and the socioeconomic status of various suburbs.

With the rise of property prices and the average home now costing more than 15 times the average income in 2020 (Australian Bureau of Statistics, 2022), Melbourne is facing a severe housing crisis. The increasing property demand of certain suburbs has resulted in an increase in the cost of living and property values, displacing many low-income families by "pricing them out of the area" (Chong, 2017). In order to slow down the gentrification of such areas, the Government may wish to take action using the information provided.

Our study will allow the Government to narrow down where to focus their efforts, as we supply information on how rising prices have led to an increase in housing insecurity. This study also provides investment opportunities, which the Government can use to improve local communities. Based on the housing prices of certain suburb areas, the Government can plan for more residential projects in certain communities that will benefit the most.

2. Methodology

The investigation began with the preprocessing of the provided datasets. Since our research question was geared towards suburbs, regular expressions (regex) were used to first filter the communities.csv dataset. This was followed by the normalisation of suburb names by converting to lowercase and removing leading or trailing spaces. The localities from houses_by_suburb were extracted and normalised as well to ultimately find common suburbs between both data sets. As a result, both data frames were merged to ensure a cohesive analysis.This was followed by a more thorough normalisation of all the data. Any NaNs found in columns with numerical, continuous data, were replaced by the median value of the column.

Once our data had been cleaned and preprocessed, we commenced with our exploratory data analysis. The main objective of this part of the exploration was to delineate relationships between the change in percentage of house prices over the years, in comparison to the metrics contained within the dataset. This involved data visualisations such as heatmaps and bar charts, and calculating mutual information. The findings from these data analyses were utilised for feature selection for the machine learning models. To find mutual information, the entropies of the metrics were calculated, as well as the percentage of house prices. The conditional entropy was then found. Afterwards, these mutual information values were normalised through the division of the minimum entropy.

Machine learning models were then employed to predict the Index of Relative Socioeconomic Disadvantage (IRSD) based on change in housing price. This was observed as it identifies which suburbs should be noted for government areas of focus to help tackle housing insecurity by way of gentrification. Linear Regression was used to map the relationship between IRSD and housing price. kNN was implemented to predict IRSD for the current year based on available data.

3. Data Exploration and Preprocessing

3.1 Data preprocessing

During the data preprocessing phase, we first normalised the names of all the suburbs in both the ‘suburbs.csv’ and ‘houses-by-suburbs.csv’ dataset. We then filtered out all common suburbs that occurred in both datasets. This step is crucial, since not all suburbs listed in the dataset are present in the housing dataset, and would cause issues when comparing the two datasets together. The two datasets were then merged together. Following this, in order to get rid of invalid values in the dataset, any spaces or blanks were first replaced with NaN values. Strings were attempted to be converted into numeric values, and if it could not be converted it was replaced with NaN values. The NaN values were then replaced with the **median** of that column. Using the median is important to maintain accuracy: NaNs as simply 0s would lead to misleading results, and we choose the median over the mean because the median is much less sensitive to outliers.

Lastly, we converted the rest of the numerical, continuous values into usable percentages through min-max scaling. Min-max scaling was chosen over binning because after looking through the factors we were going to min-max scale, we noticed that there were hardly any noticeable outliers, and we did not want to risk losing any important information by binning.

The dataset involves the factors that we thought could potentially affect housing prices throughout 2013 to 2023:

1. Travel time to the nearest public hospital
2. Distance to nearest public hospital
3. Percentage of top industry
4. Percentage of top occupation
5. Percentage of unpaid carer to person with disability
6. Unemployment rate
7. Percentages of dwellings with no internet
8. IRSD

9. Number of households with an income of less than $600/week.
10. Number of public hospitals
11. Number of private hospitals
12. Number of community health centres
13. Population density

These factors, mainly healthcare and income, were selected because they were thought to potentially impact housing prices the most. For example, accessible healthcare increases the demand and value for houses in the area because healthcare is an essential for many individuals and families. Unemployment rate and one's occupation also links to one's income and their ability to afford the area.

3.2 Heatmap

From the selection of factors, we created a heatmap to show the relationship of how each factor correlates with the changes in housing prices from 2013-2023. Through the correlation heatmap, we can observe how correlated two variables are in a linear relationship.

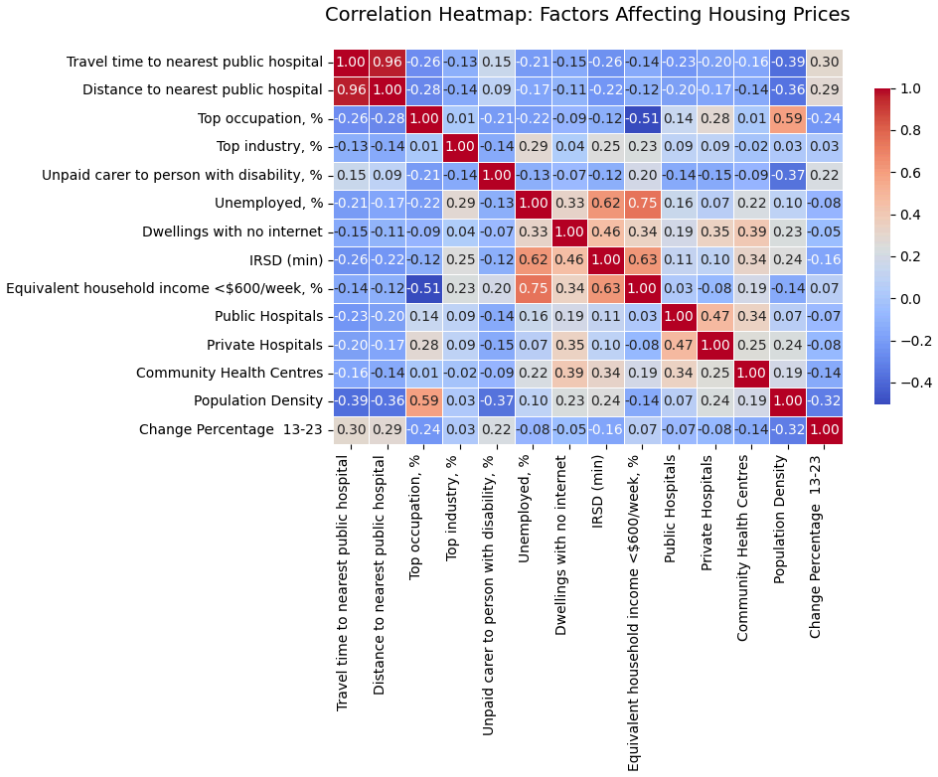


Figure 1: Heatmap: different factors affecting housing prices

In figure 1, the population density has a positive correlation of 0.59 to the percentage of top occupations in the area, indicating that areas with a higher population density will have a higher percentage of top occupations. Through the use of heatmap, it is easy to visually assess how strongly or weakly a factor is correlated to the change in housing prices.

With a heatmap, a positive correlation of 1 suggests that as a factor increases, so will the change in housing prices. Conversely, a negative correlation of -1 suggests that as certain factors increase, the change in housing prices will decrease. If there is a weak correlation or no correlation at all between factors and the housing price, the correlation will tend to 0, meaning the factor does not affect the housing price at all. Therefore, the heatmap produces an insightful look into the relationship between the key

factors that are selected with the housing price, helping us to identify the factors that have an impact on how housing prices change.

3.3 Bar plots

In addition to the heatmap, we also created another visualisation, a bar plot. This is to explore more on the relationship of how the impacts on housing price changes affect low-income families in selected areas that were the most impacted. In this section, low-income families are defined as the percentage of families with a household income of less than $600/week.

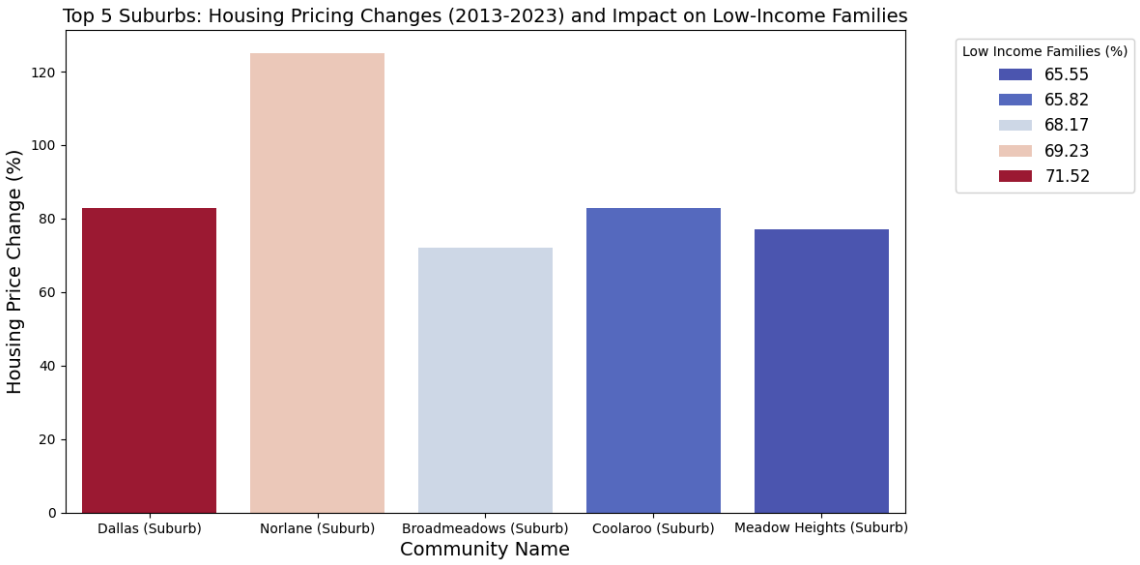


Figure 2: Barchart: Top 5 suburbs with low income families

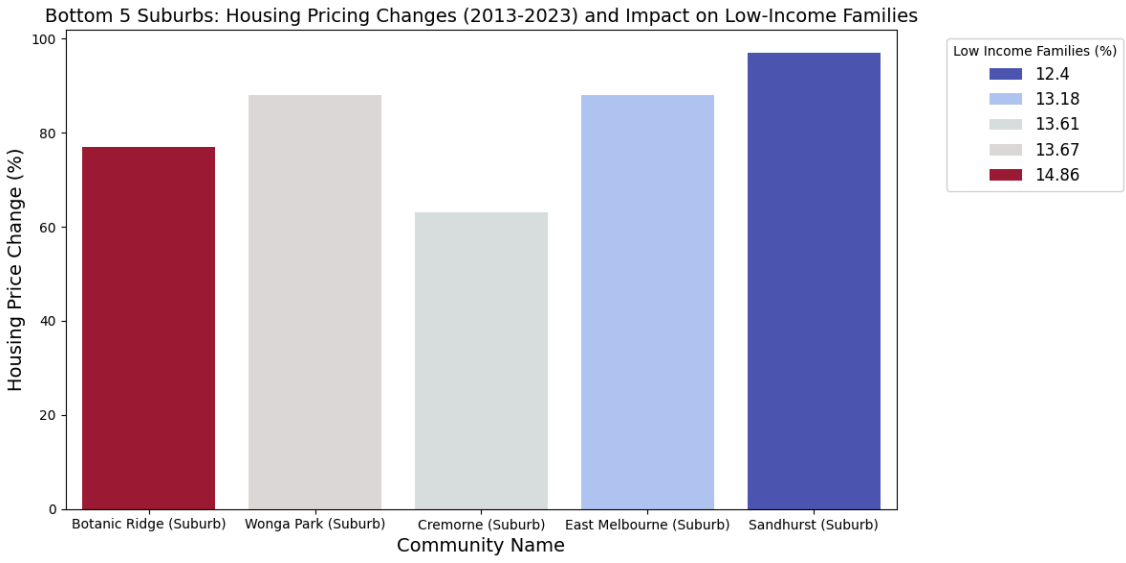


Figure 3: Barchart: Bottom 5 suburbs with low income families

The bar plot highlights the top 5 (figure 2) and bottom 5 (figure 3) suburbs that were most impacted by the housing price change between 2013 and 2023.

Each bar represents the percentage change in housing price for the given suburbs and with different colours, it indicates the proportion of low-income families. By using these visualisations, we can understand how housing price changes disproportionately have an impact on the low-income families within those areas.

3.4 Mutual information

We also utilised mutual information to quantify the relationship between two random variables. This was especially useful in the context of our investigation, providing a measurement of how much information one variable provides about another regardless of whether these variables had a linear or non-linear relationship. These values were then used for feature selection in the implementation of our machine learning models, for k-Nearest Neighbours in particular.

Factors	Mutual Information (normalised)
Travel time to nearest public hospital	0.108929
Distance to nearest public hospital	0.111065
Top occupation, %	0.122925
Top industry, %	0.112991
Unpaid carer to person with disability, %	0.115901
Unemployed, %	0.070157
Dwellings with no internet	0.072883
IRSD (min)	0.108769
Equivalent household income <$600/week, %	0.080476
Public Hospitals	0
Private Hospitals	0.08639
Community Health Centres	0.148766
Population Density	0.140032
Change Percentage 13-23	1
2023 (House Price)	0.10627

Figure 4: Table of normalised mutual information scores against Change Percentage 13-23

Following the calculations and normalizations of the mutual information scores mapped against the most recent house price available in our dataset, there are evidently relatively low mutual information scores across the board, as seen by figure 4. The strongest relationships to Change Percentage 2013-2023 seen from the table above is Community Health Centres, Population Density, and Top occupation %. These findings are helpful for the implementation of the machine learning models, especially kNN, which is optimised by feature selection.

4. Data Modelling

In this analysis, two models, linear regression and k-Nearest Neighbours, were applied to explore housing price changes and socioeconomic factors.

4.1 Linear Regression

We used a linear regression model in an attempt to explore the linear relationship between the percentage change in houses and the IRSD (min) of different suburbs. The lower the IRSD, the more economically disadvantaged an area is. In order to see if the change in price would affect the IRSD, the percentage change is set as the independent variable and the IRSD the dependent variable. The data set is split into a 80%/20% train/test set to prevent overfitting. The independent and dependent variables are then fit into the linear regression model, and a scatter plot of the data points, as well as the regression line, are plotted.

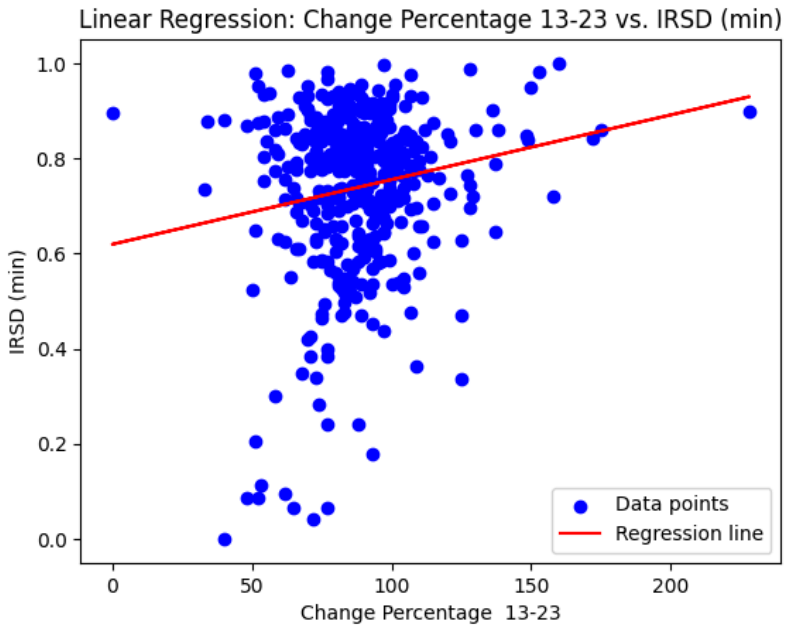


Figure 5: Linear regression: the percentage change in houses prices from 2013-2023 vs the IRSD (min)

Looking at figure 5, the data points are weakly clustered around the regression line, and the negative slope of the line itself is weak. A majority of the points are clustered around an IRSD range of more than 0.5, along the 50-100% change in housing prices, with the IRSD values remaining the same as the change in housing prices increases. This suggests a weak linear relationship between the percentage change in house prices and the socioeconomic index of the area. The points with the lowest IRSDs have a percentage change in housing prices of around 50-70%.

Since this is a regression model instead of a classification model, we used R^2 to measure the proportion of variance in the IRSD that is explained by the percentage change in housing prices. The larger the R^2 value, the stronger the relationship between the two. On the test dataset, we obtained an R^2 value of 0.0629, which is relatively low and once again suggests that the linear relationship between the IRSD and percentage change in housing prices is not very strong.

4.2 k-Nearest Neighbours

The k-Nearest Neighbours (kNN) supervised machine learning model was implemented to be able to predict housing prices for 2024, since this information was outside of the scope of the provided datasets. kNN was well suited for this particular task due to its simplicity, interpretability, and ability to handle non-linear relationships between variables. It works by finding the closest historical data points (neighbours) to the new observation and using their known values to make predictions.This approach is particularly effective for predicting changes in house prices from 2013 to 2023, as it leverages patterns observed in the historical data during that period. Key features were selected based on the upper half of the normalised mutual information scores. We used accuracy as the evaluation metric, and the model achieved an accuracy score of 0.75. This kNN model was implemented with k = 2, as this was the value that was optimised for accuracy.

The dataset was split into a 80-20 ratio between testing and training sets to train the kNN model.

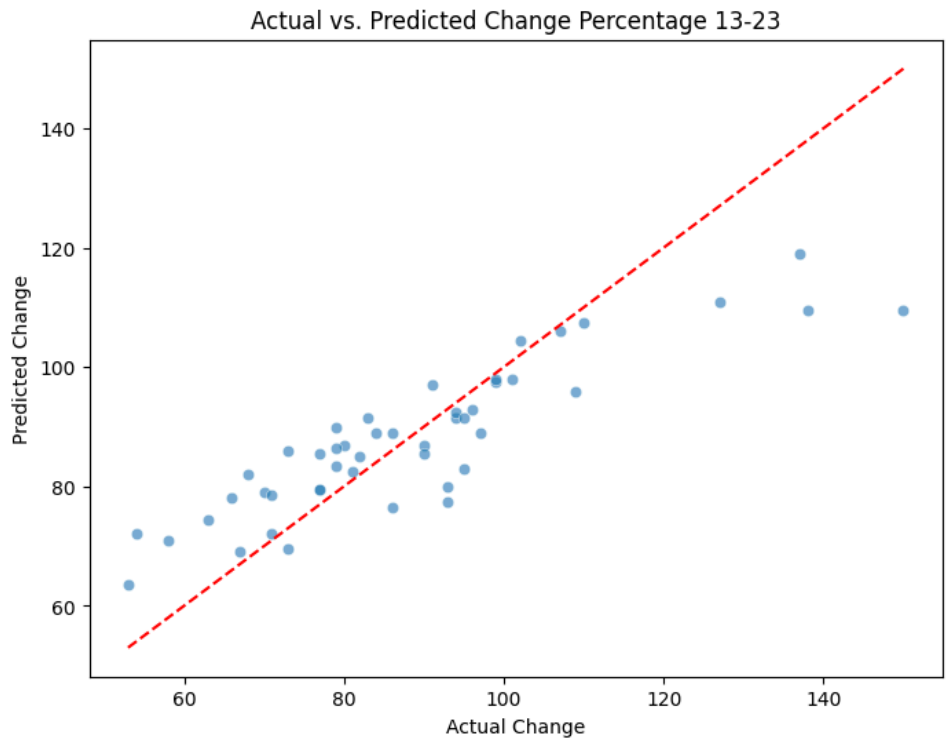


Figure 6: Actual vs Predicted Change in house price 13-23 using the kNN model

The spread of points reveals that the model generally tracks the data's trends, but there are some outliers where the model either significantly overpredicts or underpredicts the actual change percentage. These outliers may be influencing the overall accuracy of the model. Towards the left end of the graph, the points are more densely clustered. Most points are clustered between approximately the 70 to 100 range, indicating that the model performs better within this interval. Areas with fewer points suggest that there are fewer data points in those ranges. If these areas also correspond to higher or lower actual changes, it may indicate that the model struggles to predict those values accurately, possibly due to lack of data or inherent variability in the actual changes.

The clustering can help identify where the model may need refinement.The model overpredicts the values up to around the 70 mark, and underpredicts from 90 onwards. It severely underpredicts the higher values and it may be beneficial to gather more data or consider additional features that could improve the model's predictions in these specified areas.

5. Discussion and Interpretation

Our analysis and models reveal a weak relationship between the IRSD and the percentage change in housing prices, alluding to the gentrification of certain areas. Gentrification is defined as

> "a process in which a poor area (as of a city) experiences an influx of middle-class or wealthy people who renovate and rebuild homes and businesses and which often results in an increase in property values and the displacement of earlier, usually poorer residents" (Merriam-Webster)

The weak correlation between IRSD and housing prices shows us that the housing market does not align with the financial capacity of the communities it serves, exacerbating the wealth gap and inequality in gentrifying neighbourhoods. The government must take the needed action to ensure gentrification is slowed, and that housing remains affordable.

5.1 Interpretation of our data analysis

The heat map analysis helps us identify whether the factor that has been chosen is strongly or weakly correlated with how the housing price changes over time. With the heat map that was provided, we can see that the correlation between the population density to the change in percentage of housing prices from 2013-2023 is -0.32. From this correlation percentage value, the suburb with the higher percentage of population density will have a smaller increase in the housing price within this period. The negative correlation of the population density with the change percentage 13-23, it reveals that highly populated areas did not experience the same level of the price growth with the less densely populated areas. This is a significant finding because it highlights the economic downturn within these regions despite their affordability. The insight of the heat map provides a better understanding of how the housing price progressively affects the areas and populations where low-income families live in more dense areas.

The two bar graphs compare the top 5 and bottom 5 suburbs in terms of housing price changes between 2013 and 2023, using a colour scale to indicate the percentage of low-income families in each suburb. From the top 5 suburbs of low-income families (figure 2) , in suburbs such as Norlane and Dallas, housing prices have increased by 120% and 80% respectively. These suburbs also have the highest percentage of low-income families, with percentages of 69.23 in Norlane and 71.52 in Dallas. This shows that as the housing prices increase, there is still a high number of substantial low-income families. The increase in the prices of housing suggests that these areas might be undergoing gentrification: the property values are rising but the income of families stay stagnant. If the price growth continues to increase, the affordability of the suburb will decrease, potentially impacting these low-income families. On the other hand, suburbs like Sandhurst and East Melbourne are the suburbs with the smallest percentage of low-income families. As shown on the barplot in figure 3, the percentage of low income families are 12.4 and 13.18 respectively. This suggests that these suburbs contain a wealthier population, highlighting their economic stability. Despite their lower concentration of low-income families, housing price changes still change substantially, suggesting that there are other factors that affect the house pricing changes. The visualisation of the bar plots are valuable as it shows the effect of housing price changes, and how it affects different parts of the population. The government may then use this information to implement policies where they can balance the development of affordability with the placement of the low-income families in such low-income areas.

5.2 Interpretation of our models

As for our linear regression model, we obtained an R^2 value of 0.0629 as mentioned above. This is a relatively low value, suggesting that the linear relationship between the IRSD and percentage change in housing prices is not very strong. One might expect a strong correlation between the housing prices and the socioeconomic conditions in an area. The weak relationship between the two is concerning since it suggests that the housing market is not accommodating to the disadvantaged. As mentioned in the introduction, the gentrification of areas results in the displacement of low-income residents, replaced with wealthier individuals and making it appear as though the area's IRSD has not changed, when in reality it does not reflect an improvement in the quality of life of residents in the area. The Government must take action to ensure that previous residents, especially minorities and the disadvantaged, are not forced into poorer areas.

The kNN model shows a notable increase in accuracy when the features related to the IRSD and households earning <$600 per week were excluded from the implementation. Initially, with these factors included, the kNN model yielded an accuracy of around 0.71, which grew to 0.75 afterwards. This finding highlights a concerning trend, alluding to a disregard among government and real estate professionals for aligning housing prices with the socioeconomic statuses of residents of these suburbs. The low correlations apparent in the heat map and mutual information scores further support this assertion. Housing prices do not correlate as strongly with IRSD values as initially expected, indicating that the market is not accommodating to the socioeconomic realities of these communities. This disconnection further perpetuates the ongoing housing crisis, as the results demonstrate that neither the IRSD nor the measure of equivalent household income below $600 per week holds very strong relationships against the

change in house price 2013-2023. This insight emphasises the need for a more nuanced understanding of how socioeconomic factors influence housing markets and the importance of developing strategies that reflect the needs of low-income families.

5.3 Recommendations

Following this investigation, we have come to find that the government does not consider socioeconomic status of the suburb when determining house prices, which is criminal given the current housing crisis. Housing prices rising quickly leads to an increase in living costs regardless of the community's ability to afford these prices. In order to prevent the displacement of initial residents, we recommend that the government consider 3 policies that will help slow the gentrification of neighbourhoods:

1. Implementing house pricing controls
2. Increasing the number of subsidised housing
3. Increasing the support for low-income families

Besides these policies, the government may also wish to increase the quality of healthcare, education, and community services in suburbs with many low-income residents, to ensure that even if residents are displaced, their basic needs are still met. The government needs to be mindful of IRSD scores more in relation to the change in housing prices in order to avoid the gentrification of suburbs. They must acknowledge the fact that people may not be able to afford to live there, especially in suburbs with the top lowest IRSD scores, as displayed in figure 2.

We developed a Random Forest model to predict housing prices, incorporating IRSD (Index of Relative Socio-Economic Disadvantage) values to account for socio-economic factors. The model was trained on past housing prices and demographic data.



Figure 7: Actual vs Predicted Change in house price 13-23 using the Random Forest model

With an R^2 value of 0.96, this model indicates strong predictive power. The dataset was split into training/testing sets, hence, due to certain limitations, not all suburbs were included in the predictions, and thus, three out of the five suburbs outlined in figure 2 have been predicted.

Locality	Predicted Price 2024
BROADMEADOWS	592060
COOLAROO	545905
DALLAS	549390

Figure 8: Predicted price for top 5 low-income suburbs for 2024

In figure 8, the model's predictions for suburbs like Coolaroo ($530,500 in 2023), Broadmeadows ($560,000 in 2023), and Dallas ($512,500 in 2023) were notably higher than actual 2023 prices, proposing that the housing prices could rise in the near future. With the notable percentage of low-income families in these suburbs, the government must implement price controls to regulate the housing prices and prevent these areas from being priced out. The prices recommended by the model are guidelines which maintain the government and other stakeholders' investment opportunities, while bearing in mind the IRSD values in these lower income suburbs. The government must be careful in ensuring that housing prices in the future do not go over this threshold.

6. Limitations and improvement opportunities

Our analysis faced several key limitations, such as a major difference in the timeframe of the datasets used, the oversimplification of our data, and the poor results from our models.

6.1 Timeframe of datasets

One significant limitation in our analysis is that the years in the dataset do not align. In the 'houses-by-suburb.csv' dataset, it tracks the percentage change of housing prices from the years 2013 to 2023. However, the factors selected as well as the IRSD data from the 'communities.csv' dataset consists of data from only one year between 2011-2014. The data is from almost a decade ago and is outdated, thus failing to accurately capture the timeline of the housing prices, reducing the reliability of our results. Because of the difference in the timeframes of the datasets, inconsistencies arise and predictions are less accurate.

As for the improvement opportunities, we could have found external datasets that do align with the timeframe of the 'houses-by-suburb.csv' dataset. Alternatively, we could have reframed our research question to analyse how the housing prices in 2013 **alone** affect the socio-economic status of a suburb, instead of looking at the housing prices changes throughout the years.

6.2 Simplification of the data

Another limitation was in the preprocessing of our data. We simply replaced NA values with the median of that entire column. While the median is not sensitive to outliers, it is not a representation of the overall distribution and will lead to the distortion and simplification of the data. Instead, we could have used a more advanced method such as a kNN model to estimate the missing values in the data set, enabling us to preserve the complexity of the dataset.

Min-max scaling of several factors in our data set is also a limitation. We used min-max scaling because we noticed that there were hardly outliers in our particular dataset. However, for a larger dataset, min-max scaling will be used under the presumption that there are no outliers that will skew the scaling. Instead of min-max scaling, we could have potentially used robust scaling, which scales the data based on the interquartile range and is better suited for large datasets.

When choosing the factors that we thought could potentially affect housing prices, most factors chosen looked at healthcare and economical factors, as we were working under the assumption that these were the two most important factors. This limits the range of analysis we could have done, and it would have been useful to consider other factors such as education quality, public transport, and the crime rate of the area for a much more comprehensive analysis of the data.

6.3 Weak results

Lastly, the percentages of our heatmap, and the mutual information scores for our feature selection model are poor. This suggests that many factors have a weak linear relationship with the change in housing prices. One of the reasons for this is because we simply assumed that these factors were linearly correlated to one another, which is usually not the case. In reality, housing prices are influenced by many factors that interact with each other in complex, non-linear ways. For example, housing prices are affected by the current political and economical state of the world, which was not taken into account in our analysis. Additionally, the low R^2 (evaluation matrix) score of our linear regression implies that the IRSD (min) is not not as dependent on the change of housing prices as we initially thought. As mentioned in our discussion and interpretation, gentrification and government programs could play a role in changing the dynamic of the relationship between IRSD and housing prices.

In the future, non linear models might be able to better capture the non-linear relationship between different factors. For example, the decision tree model or the random forest model is much more flexible in capturing non-linear relationships between factors. The kNN model can also be further improved by using datasets with more data points at both ends of the spectrum; for suburbs with very low or high change in housing price. This would allow the model to become more accurate at these points.

Conclusion

This report has highlighted the disconnect between housing prices in suburbs and the socio-economic conditions of the residents in these suburbs. Our key findings include

- A low correlation between the factors we have selected and the change in housing price
- A very low linear relationship between IRSD and change in housing price
- Higher accuracy score when IRSD and latest housing price is omitted from feature selection in the training of the kNN model

Our investigation provides the Government with the necessary insights and resources to focus their efforts, since we have gathered information on how rising prices have led to an increase in housing insecurity. This study includes possible investment opportunities as well, which the Government can use to improve local communities. The Government can plan for more residential projects in communities that will benefit the most, based on the housing prices of certain suburb areas.

References

Australian Bureau of Statistics (May 2022) Housing Occupancy and Costs, ABS Website, accessed 2 October 2024

Chong, E. (2017). Examining the Negative Impacts of Gentrification. *Georgetown Journal on Poverty Law & Policy*. https://www.law.georgetown.edu/poverty-journal/blog/examining-the-negative-impacts-of-gentrification/

Merriam-Webster. (n.d.). Gentrification. In Merriam-Webster.com dictionary. Retrieved October 6, 2024, from https://www.merriam-webster.com/dictionary/gentrification